

Chapter 14: Obstetrical Imaging — Revision Notes

Williams Obstetrics 26e, pp. 640–701 of the PDF. High-yield numbers are in **bold**. All 12 tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 16 figures are included from the PDF.

Big picture: Ultrasound is the core imaging tool in pregnancy: it confirms dates and viability, finds and characterizes fetal, placental and fluid abnormalities, and guides procedures. Doppler adds flow information (umbilical artery for FGR, MCA for anemia). MR imaging is a problem-solving adjunct, never the screening tool.

1. Technology and Safety

Physics

- Piezoelectric crystals convert electrical energy ↔ sound waves, emitted in pulses.
- **B-mode (grayscale):** dense tissue (bone) = bright echoes; fluid = dark. **50 to >100 frames/sec** → looks real-time.
- Ultrasound = frequency **>20,000 Hz**.
- **Higher frequency = better resolution; lower frequency = deeper penetration.**

Transducer	Frequency	Use
Transvaginal	5–12 MHz	Early pregnancy (embryo is close to the probe)
Transabdominal	4–6 MHz	1st and 2nd trimesters
Transabdominal, low frequency	2–5 MHz	3rd trimester, obesity (poorer resolution)

Embryo and fetal safety

- **ALARA** = as low as reasonably achievable. Scan only for a valid medical indication.
- **No causal link** between diagnostic ultrasound and any adverse effect in human pregnancy, including **autism**.
- Machines must display two indices:
 - **Thermal index (TI):** probability of enough heating to injure. Higher with longer exams and **near bone**; theoretical risk highest in **organogenesis**.
 - Use **TIs (soft tissue) before 10 wk** and **Tlb (bone) at ≥10 wk**.
 - **Pulsed Doppler has a higher TI than B-mode.** In the 1st trimester, if needed: **TI ≤0.7** and as brief as possible.
 - Document embryonic heart rate with **M-mode, not pulsed Doppler**.
 - **Mechanical index (MI):** risk of cavitation from rarefactional pressure. Only relevant in **gas-containing tissue** → **no microbubble contrast in pregnancy**. The fetus has no gas bodies, so it is not at risk.
- **"Keepsake" imaging** for nonmedical purposes is not condoned (FDA, AIUM, ACOG). Images from medically indicated exams may be shared.

Operator safety

- Work-related musculoskeletal injury in **~70%** of sonographers: shoulder capsulitis/tendonitis, epicondylitis, carpal/cubital tunnel, neck/back strain.
- Risk factors: awkward posture, sustained static force, pinch grips, obese patients.
- Prevention: patient close, elbow near body, **shoulder abduction <30°**, thumb up; forearm parallel to the floor; back-supported chair; monitor **~15° below horizontal**; no reaching/twisting; frequent breaks.

2. Gestational Age Assessment

- Based on **certain LMP + first ultrasound measurement**. **Gestational sac size is NOT used** to date.
- If LMP is certain, the EDD uses LMP unless the discrepancy exceeds the thresholds in Table 14-1. Uncertain/unknown LMP → ultrasound sets the EDD.
- CRL = most accurate dating method**. Transvaginal, midsagittal, neutral (nonflexed) position; **mean of 3 measurements**. Accurate to **±5–7 days before 14 wk**.
- From 14⁰ wk**: BPD, HC, AC, FL. Accurate to **7–10 days before 22 wk**. EFW accuracy **within 15%**.
- First scan at ≥22 wk = suboptimally dated** → consider a repeat scan in **3–4 wk** (especially if measurements are small → possible FGR).
- IVF, fresh transfer**: EDD = retrieval/fertilization date **+266 d**. **Day-3 frozen embryo: +263 d**. **IUI: use LMP**.

Table 14-1. Assessment of Gestational Age

Gestational Age ^a	Parameter(s)	Threshold Value to Redate ^b
<9 wks	CRL	>5 d
9 to <14 wks	CRL	>7 d
14 to <16 wks	BPD, HC, AC, FL	>7 d
16 to <22 wks	BPD, HC, AC, FL	>10 d
22 to <28 wks	BPD, HC, AC, FL	>14 d
≥28 wks	BPD, HC, AC, FL	>21 d

^aBased on last menstrual period (LMP). ^bUltrasound gestational age should be used if it differs from the LMP-derived gestational age by more than the threshold value. AC = abdominal circumference; BPD = biparietal diameter; CRL = crown-rump length; FL = femur length; HC = head circumference.

Memory hook: Redating thresholds go **5 → 7 → 7 → 10 → 14 → 21 days**: the later the scan, the more slack you give the LMP.

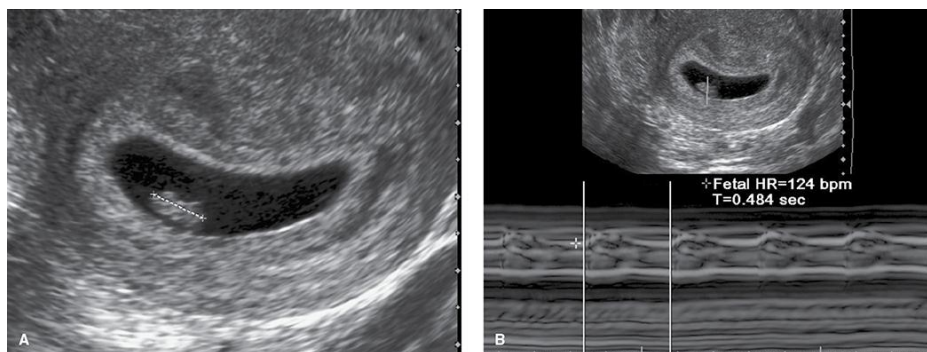


Figure 14-1. A 6-week embryo with CRL ~7 mm (A); M-mode documents cardiac activity at 124 bpm (B).

3. First-Trimester Ultrasound

Three types: (1) standard; (2) **nuchal translucency at 11–14 wk**; (3) **detailed 1st-trimester anatomy at 12–14 wk**.

- Ideal time to evaluate the **uterus, adnexa and cul-de-sac**, and to reliably diagnose anembryonic gestation, embryonic demise, ectopic pregnancy and GTD.
- Chorionicity is most accurate in the 1st trimester**. Cesarean scar pregnancy is increasingly detected now.

Key transvaginal milestones

Finding	Timing / threshold
Intrauterine gestational sac	Visible by 5 wk
Embryo with cardiac activity	By 6 wk
Embryo must be seen once MSD reaches	25 mm (otherwise anembryonic)
Cardiac motion may be seen at CRL	2 mm
Cardiac motion should be seen at CRL	7 mm
CRL <7 mm with no heartbeat	Rescan in 1 week
Parkland: demise diagnosed at	CRL ≥10 mm without cardiac motion

- Cardiac motion is documented by **video clip or M-mode**.
- New in the standard exam (when size permits): **calvarium, ventral-wall cord insertion, presence of limbs**.
- Criteria for 1st-trimester demise: Table 11-2.

Table 14-2. Indications for First-Trimester Ultrasound Examination

- Confirm an intrauterine pregnancy
- Estimate gestational age
- Confirm cardiac activity
- Diagnose/evaluate a multifetal gestation, including chorionicity and amnionicity
- Assess for certain fetal anomalies, such as anencephaly
- Measure fetal nuchal translucency, when part of an aneuploidy screening program
- Evaluate for uterine abnormalities or pelvic masses
- Evaluate for suspected ectopic pregnancy
- Evaluate for suspected gestational trophoblastic disease
- Evaluate for the cause of vaginal bleeding
- Evaluate for the cause of pelvic pain
- Serve as adjunct to embryo transfer, chorionic villus sampling, and intrauterine device localization and removal

Table 14-3. Components of Standard Ultrasound Examination by Trimester

First Trimester ^a	Second and Third Trimester
Gestational sac size, location, and number	Gestational age assessment
Embryo and/or yolk sac identification	Fetal number, including amnionicity and chorionicity of multifetal gestations
Crown-rump length	Fetal weight estimation
Gestational age assessment	Fetal anatomical survey, including documentation of technical limitations
Fetal number, including amnionicity and chorionicity of multifetal gestations	Fetal cardiac activity, documented with M-mode or 2-dimensional video clip
Embryonic/fetal cardiac activity, documented with M-mode or 2-dimensional video clip	Fetal presentation
Fetal anatomy assessment including calvarium, nuchal region, ventral wall cord insertion, and presence of limbs (depending on gestational age and fetal size)	Amniotic fluid measurement (single deepest pocket, amniotic fluid index, or qualitative assessment)
Fetal nuchal translucency assessment	Placental location, appearance, and relationship to internal cervical os ^a
Maternal uterus, adnexa, and cul-de-sac evaluation	Placental cord insertion site documentation when technically possible
	Evaluation of the maternal uterus, adnexa, and cervix ^a

^aIf a transabdominal examination is not definitive, transvaginal (or transperineal) evaluation is recommended.

Nuchal translucency (NT)

- = maximum thickness of the subcutaneous translucent area at the back of the fetal neck, measured sagittally at **11–14 wk**.
- ↑ NT → ↑ risk of **aneuploidy** and structural anomalies, especially **heart defects**. Part of 1st-trimester aneuploidy screening (Ch 17); risk calculation depends on CRL.
- **NT ≥3 mm** → indication for a **detailed anatomical survey**.

Table 14-4. Guidelines for Nuchal Translucency (NT) Measurement

- Angle of insonation is **perpendicular** to NT line
- Fetus is measured in **midsagittal plane**, with nasal tip, palate, and diencephalon visible
- Margins of NT edges are visible
- Majority of image is filled by the fetal head, neck, and upper thorax
- Fetal neck lies in a **neutral position**, not flexed or hyperextended
- **Amnion line must be separate** from the NT line
- + calipers are placed perpendicular to the fetal long axis, on the **inner borders** of the nuchal membranes, with none of the horizontal crossbar protruding into the space
- Measurement should be obtained at the **widest** NT space
- **Largest of 3** NT measurements should be used

Detailed first-trimester examination (12–14 wk)

- For at-risk pregnancies at specialized centers. **Does NOT replace** the detailed 2nd-trimester survey.
- Detection of major anomalies at NT scan: **46% low-risk, >60% high-risk** (Karim, 2017).
- **Good detection**: anencephaly, lobar holoprosencephaly, ventral wall defects.

- **Poor:** only $\frac{1}{3}$ of **major cardiac anomalies**. **None** of: microcephaly, agenesis of the corpus callosum, cerebellar anomalies, CPAM, bowel obstruction (many haven't developed yet).

4. Second- and Third-Trimester Ultrasound

- **Offer to all women at 18–22 wk** (often done before 20 wk to preserve management options).
- Women average **4–5 sonograms** per pregnancy.
- Exam types: **standard, specialized, limited**. Specialized = detailed anatomy, PAS evaluation, fetal echo, Doppler, biophysical profile.

Table 14-5. Indications for Standard Second- and Third-trimester Ultrasound Examinations

- Routine evaluation of gestational age and fetal anatomy^a
- Fetal growth evaluation or size-date discrepancy
- Fetal abnormality (follow-up evaluation)
- Amniocentesis or other procedure
- Cervical length assessment
- Multifetal gestation
- Vaginal bleeding
- Placenta previa or low-lying placenta^b
- Vasa previa^b
- Placenta accreta spectrum^b
- Placental abruption^b
- Uterine or adnexal abnormality^b
- Gestational trophoblastic disease^b
- Amnionic fluid volume abnormality^b
- Preterm rupture of membranes or preterm labor
- Inability to document fetal heart tones
- Assessment of fetal well-being
- Assessment of fetal presentation
- Adjunct to external cephalic version

^aStandard ultrasound examination should be offered in all pregnancies, ideally at 18–20 weeks' gestation. ^bIncludes evaluation of suspected cases.

Standard examination

- Fetal number, presentation, cardiac activity, biometry, fluid, placental location, cervical length and anatomy survey (Table 14-3; components in Table 15-1).
- **Twins:** number of chorions and amnions, size comparison, fluid in each sac, fetal gender.
- **AIUM 2018 updates:**
 1. Added **hands and feet**, and when feasible the **3-vessel and 3-vessel-trachea views**.
 2. Placenta–cervix relationship unclear transabdominally → **transvaginal**.
 3. Cervix abnormal/not seen → transvaginal; **cervical length must come from a transvaginal image**.
 4. **Velamentous cord insertion** → color/pulsed Doppler to exclude **vasa previa**.

Detailed ("targeted", 76811) examination

- For ↑ risk of a structural or genetic abnormality (Table 14-6). Indication-driven; **not repeated** without a new risk factor.
- Table 15-1 lists ~**70** possible components; AIUM accreditation requires **50**.

- **Most common indications: maternal obesity and age ≥ 35 .**

- Obesity odds: NTD **1.9**, ventriculomegaly 1.7, cardiac 1.3, cleft 1.2, anorectal atresia 1.5, limb reduction 1.7. One study: higher anomaly rate with obesity **only when diabetes is present**.
- Age ≥ 35 : risk is from aneuploidy. **Euploid fetuses of older mothers have no higher anomaly rate.**

Table 14-6. American Institute for Ultrasound in Medicine Indications for Detailed Second- and Third-trimester Ultrasound Examinations

Category	Indications
Prior fetus or infant with a structural or genetic abnormality	—
Current pregnancy with known or suspected fetal abnormality or growth restriction	—
Increased risk for fetal structural abnormality in current pregnancy	Teratogen exposure (Chap. 8); diabetes diagnosed before 24 weeks' gestation ; nuchal translucency ≥ 3.0 mm ; abnormal serum analyte levels (e.g., elevated alpha fetoprotein); assisted reproductive technology used to achieve conception; prepregnancy BMI ≥ 30 kg/m² ; multifetal gestation
Increased risk for fetal genetic abnormality in current pregnancy	Woman or her partner carries a genetic abnormality; maternal age ≥ 35 years at delivery ; nuchal translucency ≥ 3.0 mm; abnormal aneuploidy screening test result (Chap. 17); minor aneuploidy marker found during standard ultrasound examination
Other condition affecting the fetus	Congenital infection (Chaps. 67, 68); substance abuse; alloimmunization (Chap. 18); amnionic fluid volume abnormality
Suspected placenta accreta spectrum or associated risk factors	—

Adapted from AIUM practice parameter for detailed second- and third-trimester diagnostic obstetric ultrasound examinations, 2019.

Fetal echocardiography — indications

- Suspected cardiac structural/functional abnormality; **heart rate abnormality or arrhythmia**
- Extracardiac anomaly or **hydrops**; chromosomal abnormality
- **NT ≥ 3.5 mm**
- **IVF; monochorionic twins**
- 1st-degree relative with congenital heart defect; 1st/2nd-degree relative with a Mendelian syndrome with childhood cardiac features
- Prior fetus with **heart block + maternal anti-Ro/La**
- **Retinoid** exposure; **pregestational diabetes, PKU**

Limited examination

- Answers a specific question (number, presentation, heartbeat, fluid, placenta vs os); may include biometry but no full survey.
- Done only if a standard survey was already completed (unless emergency). Otherwise, if **≥ 18 wk**, do a standard exam.

Fetal anomaly detection

- **Standard exam: ~60%** of major anomalies. **Detailed exam in high-risk: >90%**.
- **Maternal obesity $\downarrow$ detection by 20%**. EUROCAT: **40%** detected prenatally.

Anomaly	Detection	Anomaly	Detection
Anencephaly	98%	Diaphragmatic hernia	73%
Spina bifida	89%	Gastroschisis	92%
Hydrocephaly	82%	Omphalocele	90%
Cleft lip/palate	70%	Bilateral renal agenesis	94%
Hypoplastic left heart	88%	Posterior urethral valves	80%
Transposition of great vessels	69%	Limb reduction / clubfoot	60% / 60%

- **Poorly detected:** microcephaly, choanal atresia, cleft palate, Hirschsprung disease, anal atresia, congenital skin disorders.
- Every exam should include a frank discussion of its **limitations**.

Three-dimensional ultrasound

- A stored volume is rendered in any plane (static); 4-D = rapid reconstruction that looks real-time.
- Helps with **face, skeleton, tumors, some NTDs**, but **no better overall detection** than 2-D. ACOG: proof of clinical advantage is generally lacking.

5. Placenta and Cervix

Placenta previa and low-lying placenta

- Placental location vs cervix can be assessed accurately by **~16 wk**; use transvaginal if transabdominal is limited.
- **Placenta previa:** placenta **overlies or reaches the margin** of the internal os.
- **Low-lying placenta:** inferior edge **within 2 cm** of the internal os but not reaching it.
- **Follow-up at ~32 wk** (transvaginal if needed); if it persists → **repeat at 36 wk**.

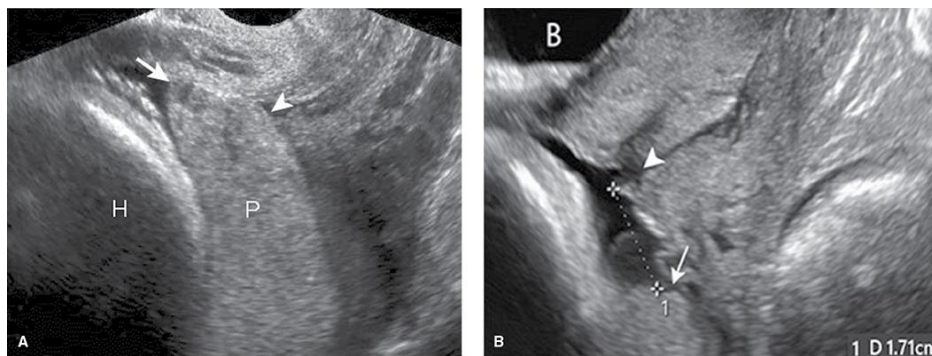


Figure 14-2. Transvaginal views: placenta previa, edge covering the internal os (A); low-lying placenta, edge within 2 cm of the os (B).

Cord insertion and vasa previa

Entity	Definition
Marginal (battledore)	Cord inserts at the placental edge or within 2 cm of the margin
Velamentous	Cord inserts into the membranes ; vessels run unprotected by Wharton jelly
Vasa previa	Fetal vessels in membranes over or within 2 cm of the cervix (velamentous cord or vessels linking a succenturiate lobe)

- Diagnosis: **transvaginal color Doppler**; pulsed Doppler of the vessel shows a **fetal heart rate** → confirms. Detailed follow-up at **32 wk**.

Placenta accreta spectrum (PAS)

- Accreta / increta / percreta = invasion **onto / into / through** the myometrium.
- Evaluate transabdominally + transvaginally, ± color/power Doppler, **bladder partially filled**.
- Ultrasound **sensitivity 91 / 93 / 89%**, **specificity 97 / 98 / 99%** (accreta / increta / percreta).

Five sonographic criteria:

1. **Placental lacunae** (vascular spaces ± prominent flow)
 2. **Retroplacental myometrial thinning**, smallest thickness **<1 mm** (loss of the retroplacental clear space)
 3. **Disrupted bladder–uterine serosal interface** (irregular echogenic line)
 4. **Bridging vessels** from placenta to bladder–serosal interface (color Doppler)
 5. **Placental bulge** distorting the uterine contour; a focal exophytic mass in some percretas
- Accuracy depends on the number of findings and risk factors (previa, number of prior cesareans). **Previa + prior cesarean + any finding → detailed evaluation**. Serial scans help, especially in the 3rd trimester.

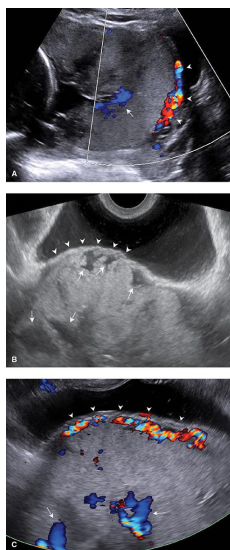


Figure 14-3. PAS signs: bridging vessels and large lacunae (A); bulge toward the bladder with irregular lacunae (B); disrupted bladder–serosal interface with myometrium <1 mm (C).

Cesarean scar pregnancy (CSP)

- Implantation in a prior hysterotomy scar; often a **precursor of PAS**.
- Sac **low and anterior**, resting on the scar or filling the **"niche"** (myometrial pocket defect).
- Placental sonolucencies (precursors of lacunae); anterior trophoblast-to-serosa distance **<3 mm**; prominent vascularity at the scar; sac may bulge toward the bladder. (Management: Ch 12.)

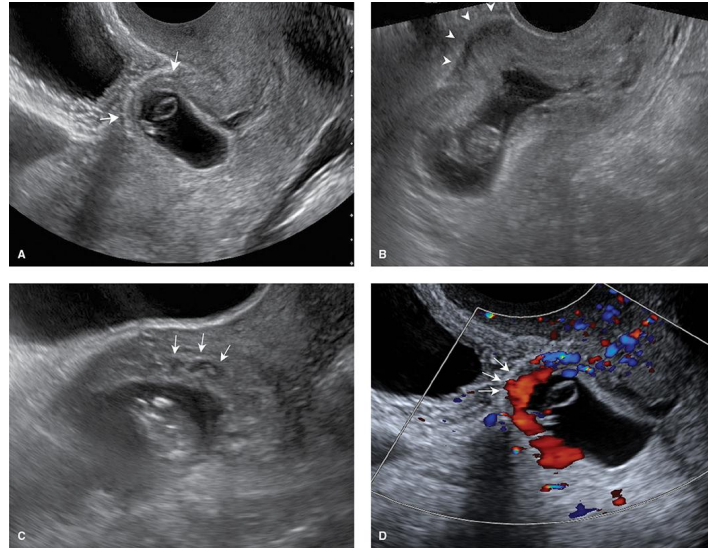


Figure 14-4. Cesarean scar pregnancy: trophoblast filling the niche, bulge toward the bladder, sonolucencies and <1 mm myometrium with scar vascularity.

Cervical length

- Transabdominal views are limited (habitus, fetal part) and a full bladder or probe pressure **artificially lengthens** the cervix.
- **Only transvaginal cervical length is used for clinical decisions**; measure at ≥ 16 wk.
- Short cervix $\rightarrow$ $\uparrow$ preterm birth risk, proportional to shortening, especially with a **prior preterm birth** (Ch 45).
- **Funneling** = membranes protrude into a dilated upper canal. **Not an independent predictor**, but linked with shortening. **Measure the cervix distal to the funnel** (the funnel base is the functional internal os).
- Dilated cervix (insufficiency) $\rightarrow$ membranes prolapse into the vagina $\rightarrow$ **hourglass** appearance.
- **Sludge** = particulate debris near the internal os; **further $\uparrow$ preterm birth risk** in at-risk women.

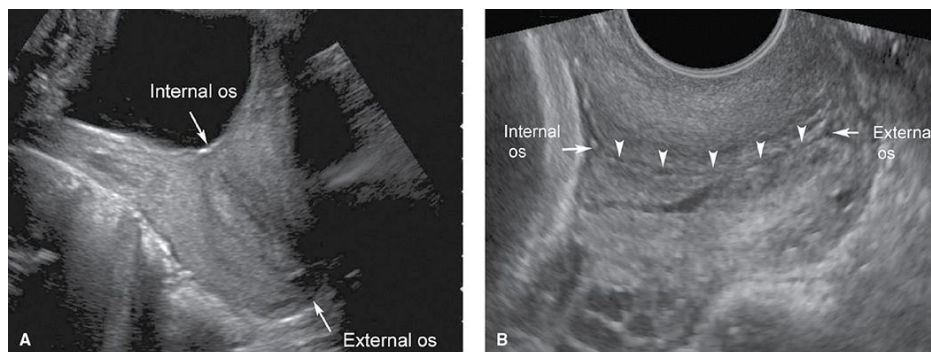


Figure 14-5. Cervix transabdominally (A) vs transvaginally (B): only the transvaginal length is used for decisions.

Table 14-7. Criteria for Transvaginal Evaluation of the Cervix

Imaging the cervix

- Maternal **bladder should be empty**.
- Transducer is inserted under real-time observation, identifying midsagittal plane, internal os, and then external os, while keeping the internal os in view.
- Internal os, external os, and entire endocervical canal should be visible. The internal os may appear as a small triangular indentation at the junction of the amniotic cavity and endocervical canal.
- Image is enlarged so that the cervix fills approximately **75% of the screen**.
- Anterior and posterior width of the cervix should be approximately equal.
- Transducer is pulled back slightly until the image begins to blur, ensuring that pressure is not placed on the cervix, then inserted only enough to restore a clear image.

- Images should be obtained with and without fundal or suprapubic pressure, to assess for dynamic change or shortening on real-time imaging.

Measuring the cervix

- Calipers are placed at the point where anterior and posterior walls of cervix meet.
- Endocervical canal appears as a faint, linear echodensity.
- If canal has a curved contour, a straight line between the internal and external os will deviate from the path of the endocervical canal.
- If midpoint of the line between the internal and external canal deviates by ≥ 3 mm from the endocervical canal, measure the cervical length in **two linear segments**.
- Funneling, sludge (debris), or dynamic change is noted.
- At least **three separate images** are measured during a period of at least **3 minutes** to allow for dynamic change. Visualization of cervical shortening on real-time imaging, with or without fundal or suprapubic pressure, raises preterm birth risks.
- **Shortest** cervical length image that meets all criteria should be used.

Adapted from AIUM, 2018a; Iams, 2013.

Memory hook: NT uses the **largest** of 3 measurements; cervical length uses the **shortest** of 3. Both err toward detecting risk.

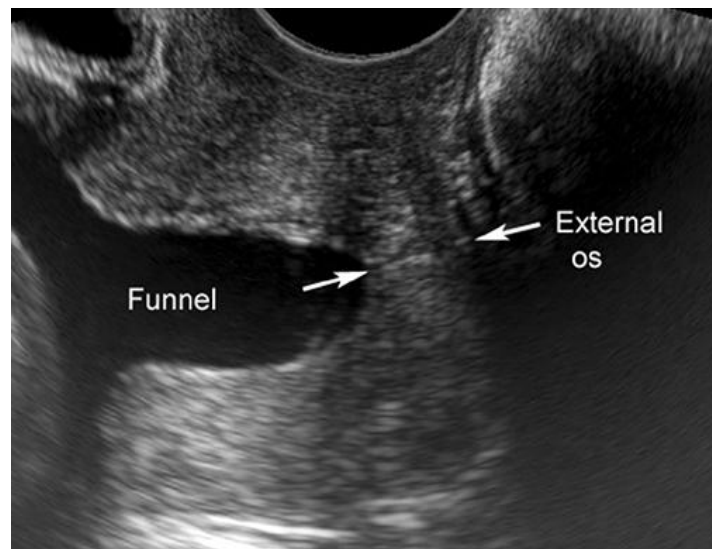


Figure 14-6. Funneling: the cervical length is measured from the funnel tip (functional internal os) to the external os, excluding the funnel.

6. Amnionic Fluid

Physiology

- Roles: **lung growth** (fetal breathing), **GI development** (swallowing), space for movement, protection from **cord compression** and trauma, bacteriostatic.
- Early fluid = transudate (like extracellular fluid): **transmembranous** (across amnion), **intramembranous** (across fetal vessels on the placental surface), **transcutaneous** (across fetal skin, until **keratinization at 22–25 wk**).
- **Fetal urine is the main source from the 2nd half** of pregnancy; it becomes major only in the 2nd trimester → lethal renal anomalies may not cause severe oligohydramnios until **after 18 wk**.
- **Swallowing = main resorption route (500–1000 mL/d)**. The whole volume is recirculated daily at term. Impaired swallowing (CNS or GI obstruction) → hydramnios.
- **Osmolality:** plasma ~ 280 , amnionic fluid ~ 260 mOsm/L (hypotonic) → drives **up to 400 mL/d** intramembranous flow into fetal vessels. Maternal dehydration $\uparrow$ maternal osmolality → fluid moves fetus → mother.

Table 14-8. Amnionic Fluid Volume Regulation in Late Pregnancy

Pathway	Effect on Volume	Approximate Daily Volume (mL)
Fetal urination	Production	1000
Fetal lung fluid secretion	Production	350
Fetal swallowing	Resorption	750
Intramembranous flow across fetal vessels on the placental surface	Resorption	400
Transmembranous flow across amnionic membrane	Resorption	Minimal

Data from Magann, 2011; Modena, 2004; Moore, 2010.

Volume across gestation

- **30 mL at 10 wk → 200 mL at 16 wk → 800 mL by the mid-3rd trimester.**
- Dye dilution: **~400 mL at 22–30 wk, 800 mL until 40 wk, then ↓ 8% per week.**

Semi-quantitative assessment

- Measured at every 2nd/3rd-trimester scan by **single deepest pocket (SDP)** or **AFI**. Subjective assessment alone is not recommended.
- **SDP**: sagittal, transducer perpendicular to the floor; pocket **≥1 cm wide**, no fetal parts or cord (check with color Doppler).
 - **Normal >2 and <8 cm** (≈ 3rd and 97th percentiles).
 - Used in **twins/multifetal** gestations (each sac) and in the **biophysical profile (>2 cm)**.
- **AFI**: sum of the deepest pocket in each of **4 quadrants**.
 - **Normal >5 and <24–25 cm**. Mean **12–15 cm** from 16 to 40 wk (Moore).
 - Rule of thumb: **AFI ≈ 3 × SDP**. Intra/interobserver variability <1 cm on average (2 SD up to 5–7 cm).

	Oligohydramnios	Normal	Hydramnios
SDP	<2 cm	2–8 cm	≥8 cm
AFI	<5 cm	5–24 cm	≥24–25 cm

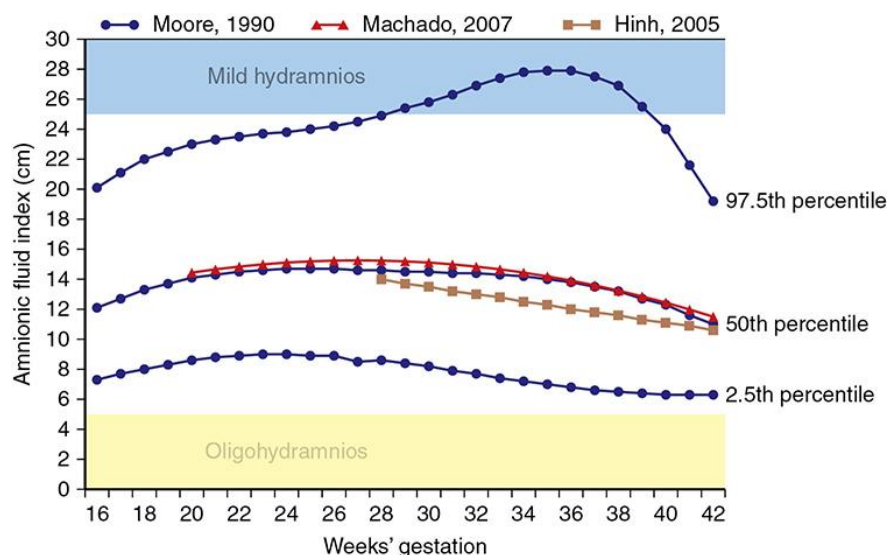


Figure 14-7. AFI nomogram: mean 12–15 cm from 16–40 wk; AFI <5 cm is below the 2.5th percentile throughout; shaded bars mark oligo- and hydramnios thresholds.

7. Hydramnios

- Complicates **1–2%** of singletons.

Severity	AFI	SDP
Mild	25–29.9 cm	8–9.9 cm
Moderate	30–34.9 cm	10–11.9 cm
Severe	≥35 cm	≥12 cm

- **Mild = ~2% of cases**, often idiopathic and benign. Severe is far more likely to have a cause and consequences.



Figure 14-8. Severe hydramnios: a pocket >15 cm, with an AFI of nearly 50 cm.

Causes

- **Fetal anomaly / genetic syndrome ~15%; diabetes 15–20%.**
- Less common: congenital infection (**syphilis, CMV, toxoplasmosis, parvovirus**), red cell alloimmunization, **placental chorioangioma**.
- **Hydrops** often coexists; mechanism is often a **high-output state** (classic example: **severe fetal anemia**).
- **Any hydramnios → detailed ultrasound.** If an anomaly is found, aneuploidy risk is significantly ↑.
- **Diabetes:** maternal hyperglycemia → fetal hyperglycemia → **fetal osmotic diuresis**. AFI correlates with amniotic fluid glucose. Repeat GDM screening is **not** helpful if the 2nd-trimester GTT was normal.
- **Twins:** hydramnios in **18%** (MC and DC alike). **MC: hydramnios in one sac + oligohydramnios in the other = TTTS**; isolated hydramnios in one sac may precede it.
- **Idiopathic** = diagnosis of exclusion; **up to 70%** of cases, **mild in ~80%**, resolves in **>1/3**.

Table 14-9. Hydramnios: Prevalence and Associated Etiologies—Values in Percent

	Golan (1993) n = 149	Many (1995) n = 275	Biggio (1999) n = 370	Dashe (2002) n = 672	Pri-Paz (2012) n = 655
Prevalence	1	1	1	1	2
Amnionic fluid index					
Mild 25–29.9 cm	—	72	—	66	64
Moderate 30–34.9 cm		20		22	21
Severe >35 cm		8		12	15
Etiology					
Idiopathic	65	69	72	82	52
Fetal anomaly ^a	19	15 ^a	8	11 ^a	38 ^a
Diabetes	15	18	20	7	18

^aA significant correlation was identified between severity of hydramnios and likelihood of an anomalous infant. The table prints "Severe >35 cm"; the text defines severe as ≥ 35 cm.

Table 14-10. Selected Anomalies and Mechanism for Hydramnios

Mechanism	Anomaly Examples
Impaired swallowing (CNS)	Anencephaly; hydranencephaly; holoprosencephaly
Impaired swallowing (craniofacial)	Cleft lip/palate; micrognathia
Tracheal compression or obstruction	Neck venolymphatic abnormality; CHAOS ^a
Thoracic etiology (mediastinal shift)	Diaphragmatic hernia ^a ; cystic adenomatoid malformation ^a ; pulmonary sequestration ^a
High-output cardiac state	Ebstein anomaly ^a ; tetralogy of Fallot with absent pulmonary valve ^a ; thyrotoxicosis ^a
Functional cardiac etiology	Cardiomyopathy, myocarditis ^a
Cardiac arrhythmia	Tachyarrhythmia ^a : atrial flutter, atrial fibrillation, supraventricular tachycardia; bradyarrhythmia ^a : heart block
GI obstruction	Esophageal atresia; duodenal atresia
Renal-urinary	Ureteropelvic junction obstruction ("paradoxical hydramnios"); Bartter syndrome
Neurological or muscular etiology	Arthrogryposis, akinesia sequence; myotonic dystrophy
Neoplastic etiology	Sacroccoccygeal teratoma ^a ; mesoblastic nephroma ^a ; placental chorioangioma ^a

^aPoses risk for hydrops. CNS = central nervous system; CHAOS = congenital high-airway obstruction sequence; GI = gastrointestinal.

Degree vs anomaly risk (Parkland, Dashe 2002)

Hydramnios	Anomalous neonate	Anomaly at birth despite normal targeted scan
Mild	~8%	1–2%
Moderate	~12%	1–2%
Severe	>30%	10%

- Anomaly found after delivery: **9%** neonatal period → **28%** by 1 year.

Management

- Severe hydramnios with **preterm labor or maternal respiratory compromise** → **amnioreduction**.
 - 18- or 20-gauge needle**; remove **1000–2000 mL over 20–30 min**; aim for the **upper normal** range.
 - May be repeated **weekly or twice weekly**.
 - In 138 cases needing amnioreduction: GI malformation **20%**, chromosomal/genetic **~30%**, neurological **8%**, idiopathic only **20%**. First procedure at **31 wk**, median delivery **36 wk**.

Outcomes

- Preterm birth, placental abruption, uterine dysfunction in labor, PPH.**
 - Abruption follows **rapid decompression** (membrane rupture or amnioreduction), sometimes days–weeks later.
 - Overdistension → **atony** → **PPH**.
- With an identified cause, severity correlates with preterm delivery, SGA and perinatal death. **Idiopathic hydramnios is generally NOT linked to preterm birth.**
- Idiopathic:** birthweight **>4000 g in ~25%** (bigger fetus → more urine); **cesarean 35–55%**.
- Stillbirth at **37 wk ~7× higher** with hydramnios (California data); risk compounded with **FGR**.

8. Oligohydramnios

- Complicates **1–2%** of pregnancies. **AFI <5 cm or SDP <2 cm**. AFI 5 cm is below the 2.5th percentile throughout the 2nd and 3rd trimesters.
- AFI labels more women as oligohydramnios than SDP, without better outcomes** (Kehl 2016: **10% vs 2%** → more inductions, same neonatal outcome).
- TTTS evaluation:** SDP **<2 cm** defines oligohydramnios.
- Anhydramnios** = no measurable pocket.

Causes

Timing / mechanism	Examples
No fetal urine (renal) — kidneys main source by 18 wk	Bilateral renal agenesis , bilateral MCDK, unilateral agenesis + contralateral MCDK, infantile ARPKD
Bladder outlet obstruction	Lower urinary tract obstruction
Complex GU anomalies	Persistent cloaca, sirenomelia
Drugs	ACE inhibitors, ARBs (2nd/3rd tri → fetal hypotension, renal ischemia, anuric renal failure; skull hypoplasia, limb contractures); NSAIDs (↓ fetal urine, ductus arteriosus constriction)
Late 2nd / 3rd trimester	Uteroplacental insufficiency (placental abnormality, preeclampsia, vascular disease); CAOS (chronic abruption–oligohydramnios sequence, usually with FGR)

- Oligohydramnios **before 20–22 wk** → **pulmonary hypoplasia**; extremely poor prognosis unless fetal therapy is possible.
- Workup: **first exclude ruptured membranes**, then a detailed ultrasound (especially in the 2nd trimester).

Management

- <36 wk with normal anatomy and growth** → **expectant management + fetal surveillance**.
- Deliver at 36⁰–37⁶ wk** (ACOG) — late-preterm/early-term fetal risks outweigh prematurity risks.
- Oral/IV maternal **hydration** ↑ **AFI**, but outcome benefit is unclear.

Outcomes

- AFI <5 cm at **24–34 wk** → ↑ perinatal morbidity and mortality (Table 14-11).
- Meta-analysis: **2×** cesarean for nonreassuring fetal status; **5×** Apgar <7 at 5 min.

Table 14-11. Pregnancy Outcomes in Women Diagnosed with Oligohydramnios between 24 and 34 Weeks' Gestation at Parkland Hospital

Factor	AFI ≤5 cm (n = 166)	AFI 8 to 24 cm (n = 28,185)	p Value
Major malformation	42 (25)	634 (2)	<.001
Stillbirth	8 (5)	133 (<1)	<.001
Gestational age at delivery ^a	35.1 ± 3.3	39.2 ± 2.0	<.001
Preterm birth, spontaneous ^a	49 (42)	1698 (6)	<.001
Preterm birth, indicated ^a	23 (20)	405 (2)	<.001
Cesarean delivery for nonreassuring fetal status ^a	10 (9)	1083 (4)	<.001
Birthweight <10th percentile ^a	61 (53)	3388 (12)	<.001
Birthweight <3rd percentile ^a	43 (37)	1130 (4)	<.001
Neonatal death ^a	1 (1)	24 (<1)	<.001 ^b

Data expressed as No. (%) and mean ± standard deviation. ^aAnomalous infants excluded. ^bThis difference was no longer significant after adjustment for gestational age at delivery. Data from Petrozella, 2011.

Borderline oligohydramnios (borderline AFI)

- **AFI 5–8 cm** (8 cm is <5th percentile in the mid-3rd trimester). Controversial.
- **No** ↑ hypertension, stillbirth or neonatal death vs AFI >8 cm (Petrozella); ↑ FGR in one study, but not preterm birth or NICU admission.
- **Insufficient evidence to support surveillance or delivery.**

9. Doppler

Principles

- **Doppler shift:** reflected frequency shifts in proportion to the velocity and direction of the moving target (RBCs).
- **Angle of insonation (θ)** must be close to **0°**; error grows as θ increases. **Ratios** cancel out cos θ.
- **S/D ratio** = peak systolic / end-diastolic flow → downstream **impedance**.
- **Continuous-wave:** two crystals (transmit + receive); used in M-mode; **cannot image individual vessels**.
- **Pulsed-wave:** one crystal; allows vessel localization and **color-flow mapping**. **Red** = toward the transducer, **blue** = away.

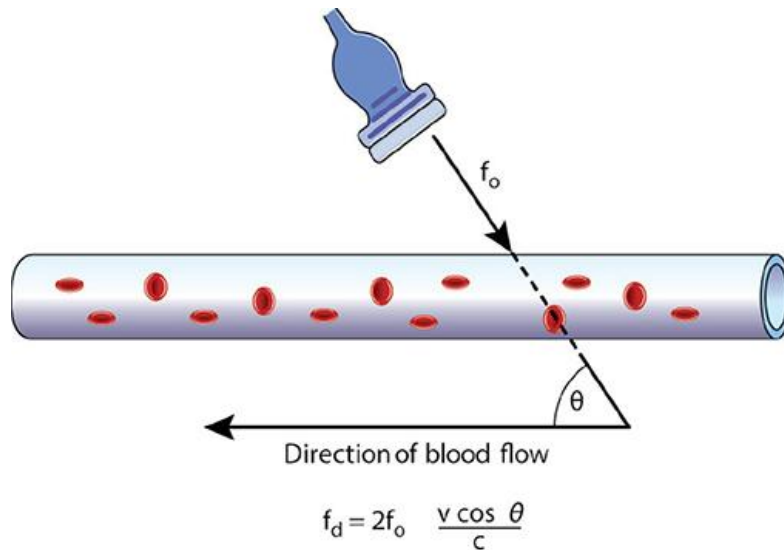


Figure 14-9. Doppler equation: $f_d = 2f_o (v \cos \theta)/c$. Accuracy depends on an insonation angle θ near zero.

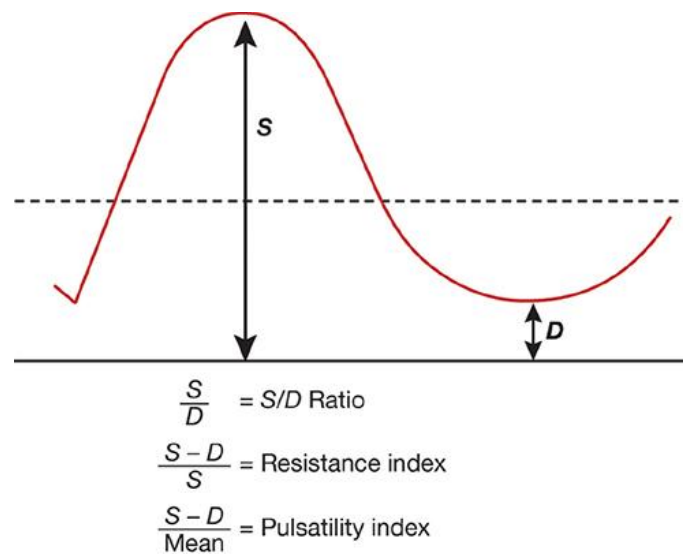


Figure 14-10. Waveform indices: S/D ratio = S/D ; resistance index = $(S - D)/S$; pulsatility index = $(S - D)/\text{mean}$.

Umbilical artery

- Normally **forward flow throughout the cardiac cycle**; diastolic flow $\uparrow$ with gestation (placental impedance $\downarrow$).
- **$S/D \leq 4.0$ after 20 wk, <3.0 after 30 wk, ~ 2.0 at term.**
- More end-diastolic flow at the **placental** end, so AEDF/REDF appears **first at the fetal ventral-wall end**. Sample a **free loop** (ISUOG); near the ventral wall may improve reproducibility when flow is low.
- **Abnormal = $S/D > 95\text{th percentile}$ for GA.**
- **Reversed end-diastolic flow $\leftrightarrow >70\%$ obliteration** of small muscular arteries in tertiary stem villi.
- **Only proven use: managing FGR** (improves outcome). **Not** for other indications and **not** for FGR screening.
- Abnormal UA Doppler $\rightarrow$ **detailed anatomy scan** (anomalies and aneuploidy are common).

Ductus arteriosus

- Used to monitor **indomethacin/NSAID** exposure, especially in the **3rd trimester**.
- Constriction $\rightarrow \uparrow$ pulmonary flow $\rightarrow$ reactive arteriolar hypertrophy $\rightarrow$ **fetal pulmonary hypertension**.
- NSAIDs $\uparrow$ odds of ductal constriction **15-fold**; risk mostly with use **>72 h**. Stop the drug if constriction is seen; usually **reversible**.

Uterine artery

- Uterine flow **50 mL/min early** → **500–750 mL/min at term**. Normal: high diastolic flow, turbulent.
- **↑ resistance and a diastolic notch** → later gestational hypertension, preeclampsia, FGR (and superimposed preeclampsia in chronic hypertension when impedance ↑ at 16–20 wk).
- **Low predictive value; not recommended for screening** in high- or low-risk pregnancies.

Middle cerebral artery (MCA)

- **Primary noninvasive test for fetal anemia** (alloimmunization surveillance); has replaced amniocentesis.
- Flow runs nearly "head-on" to the probe → accurate velocity. Axial view at the skull base, **within 2 mm of the internal carotid origin**; θ near 0, **$\leq 30^\circ$ angle correction**.
- Anemia → ↑ cardiac output + ↓ viscosity → **↑ peak systolic velocity (PSV)**.
- **MCA PSV ≥ 1.50 MoM** → **moderate or severe anemia** (Mari, 2000).
- **"Brain-sparing"** in hypoxemia (↑ end-diastolic MCA flow) is a misnomer: it signals ↑ perinatal morbidity/mortality.
- **Cerebroplacental ratio** = MCA PI / UA PI. PORTO: ratio **<1** → **11-fold** risk of adverse outcome in EFW <10 th percentile. **Not standard** for timing delivery in FGR.

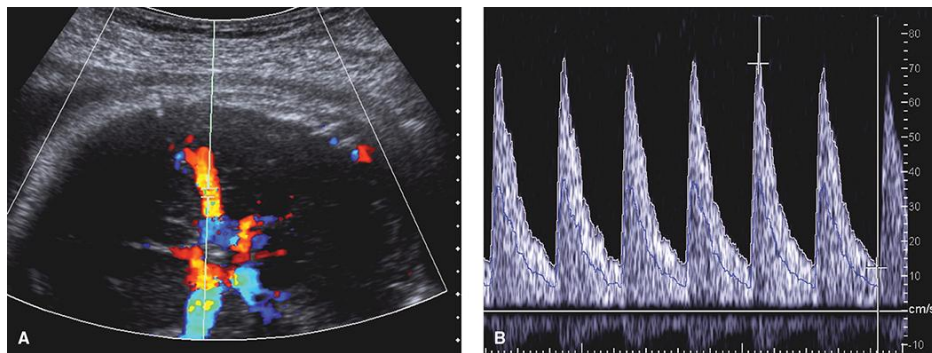


Figure 14-11. MCA sampled at the circle of Willis (A); PSV >70 cm/sec at 32 wk = severe anemia from Rh alloimmunization (B).

Ductus venosus

- Shunts oxygenated umbilical venous blood to the **IVC**, bypassing the liver. Imaged near the diaphragm.
- Biphasic, normally **forward flow throughout**: ventricular systole peak, diastolic filling peak, then the **a-wave** nadir (atrial contraction).
- Proposed FGR progression: **UA** → **MCA** → **ductus venosus**. Severe FGR with cardiac dysfunction → a-wave **decreased** → **absent** → **reversed**, plus **pulsatile umbilical venous flow**.
- **Not recommended** for routine FGR management (ACOG, SMFM); has not improved outcomes.

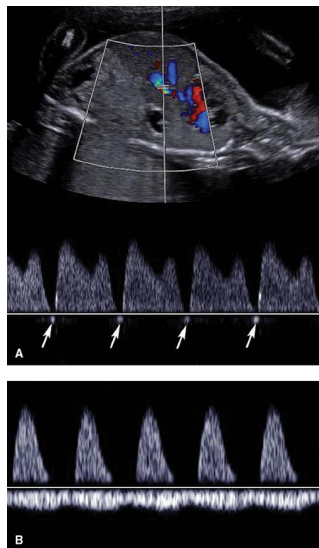


Figure 14-12. Venous Doppler abnormalities in severe FGR: reversed a-wave in the ductus venosus (A); pulsatile umbilical vein flow (tricuspid regurgitation) with absent UA end-diastolic flow (B).

10. Magnetic Resonance Imaging

Principles and use

- Excites hydrogen ions using **1.5–3 Tesla** magnets; **<1 sec per slice**.
- Simple fluid (amniotic fluid, urine) is **bright on T2, dark on T1**.
- Better than ultrasound when limited by **bone, obesity, oligohydramnios or an engaged head**.
- Downsides: not portable, time-consuming, limited to referral centers.
- **Ultrasound remains the screening modality**; MR is for **problem solving** (diagnosis, counseling, treatment, delivery planning). Parkland: MR in **15%** of pregnancies with major fetal anomalies.

Table 14-12. Fetal Conditions for Which Magnetic Resonance Imaging May Be Indicated^a

Region	Conditions
Brain and spine	Agensis of the corpus callosum; cavum septum pellucidum abnormalities; cephalocele; cerebral cortical malformation or migrational abnormalities; family history conferring a risk for brain anomaly; hemorrhage; holoprosencephaly; hydranencephaly; infarctions; monochorionic twin pregnancy complications; neural-tube defects; posterior fossa abnormalities; sacral agenesis (caudal regression); sacrococcygeal teratoma; sirenomelia; solid or cystic masses; vascular malformations; ventriculomegaly; vertebral anomalies
Skull, face, and neck	Facial clefts; goiter; hemangiomas; teratomas; venolymphatic malformations; other abnormalities with potential airway obstruction
Thorax	Bronchogenic cyst or congenital lobar overinflation; congenital cystic adenomatoid malformation; diaphragmatic hernia; effusions; extralobar pulmonary sequestration; esophageal atresia; mediastinal masses; evaluation of pulmonary hypoplasia secondary to diaphragmatic hernia, oligohydramnios, chest mass, or skeletal dysplasia
Abdomen, pelvis, and retroperitoneum	Abdominopelvic cystic mass; bowel anomalies (anorectal malformations, complex obstructions); complex genitourinary anomalies (bladder outlet obstruction syndromes, bladder exstrophy, cloacal exstrophy); renal anomalies with oligohydramnios; tumors (sacrococcygeal teratoma, neuroblastoma, hemangioma, suprarenal or renal masses)
Complications of monochorionic twins	Assess morbidity after death of a monochorionic co-twin; determine vascular anatomy prior to laser treatment; evaluate conjoined twins
Fetal surgery assessment	Anomalies for which fetal surgery is planned; fetal brain anatomy before and after surgical intervention

^aIn some cases, magnetic resonance imaging is indicated only if the anomaly is suspected but cannot be adequately characterized sonographically, which is assessed on a case-by-case basis. Summarized from ACR-SPR practice parameter for fetal MRI, 2020.

Safety

- **No known adverse fetal effects without contrast**; no ionizing radiation. **1.5T and 3T are both safe.**
- Heating is reflected by the **specific absorption rate (SAR)** (FDA-regulated); follow ALARA.
- SPR recommends **3T** (better signal-to-noise); Parkland prefers **1.5T** for diagnosis (better field homogeneity), 3T for functional imaging.
- **Gadolinium**: crosses to the fetus, is excreted into amniotic fluid and re-swallowed; the longer it stays, the more toxic free **Gd³⁺** dissociates. Linked to slightly ↑ **rheumatological, inflammatory and infiltrative skin conditions** in children; nephrogenic systemic fibrosis in adults with renal disease.
 - **Not recommended in pregnancy without a life-threatening indication.**

Technique

- Safety questionnaire first (metal implants, pacemakers).
- **Claustrophobia <1%** → single oral dose **diazepam 5–10 mg or lorazepam 1–2 mg**.
- Supine or **left lateral decubitus**; torso coil (body coil for large habitus).
- Sequence: 3-plane localizers → **T2 fast acquisition (SSFSE / HASTE / RARE), 7-mm slices, no gap** → **fast T1 (SPGR)** → targeted orthogonal T2 at **3–5 mm**.
- **T1-bright**: subacute hemorrhage, fat, **liver, meconium**. **STIR / fat-sat T2** separate lesions with similar water content (e.g., thoracic mass vs lung). **Diffusion-weighted imaging**: restricted diffusion in ischemia, cellular tumor, clot.

Fetal anatomy evaluation

- **~95%** of ISUOG components seen at 30 wk (**aorta and pulmonary artery hardest**); **99%** of non-cardiac anatomy.

Central nervous system

- **T2:** CSF bright → posterior fossa, midline and cortex detail. **T1:** hemorrhage. Biometry matches ultrasound (nomograms for corpus callosum and vermis).
- Best yield **after 24 wk**. Adds CNS findings or changes the diagnosis in **~50%**; changes prognosis/management in **20%**.
- Uses: gyration/sulcation, **corpus callosum agenesis/dysgenesis**, migrational abnormalities, **septo-optic dysplasia** (absent septum pellucidum, hypoplastic optic tracts), **intracranial hemorrhage** (extent and timing), congenital infection.
- **ICH risk factors:** atypical ventriculomegaly, suspected **NAIT**, severe TTTS, or **demise of a monochorionic co-twin**.

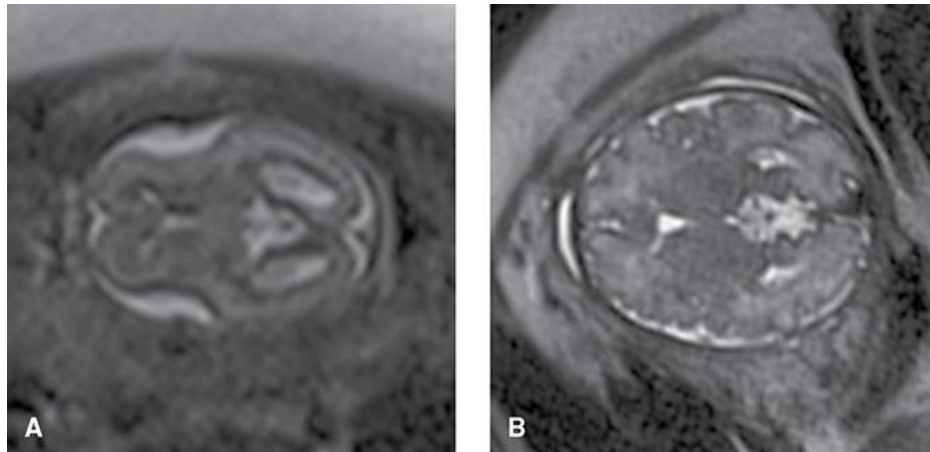


Figure 14-13. Normal gyration and sulcation: smooth brain at 23 wk (A) vs folded cortex at 33 wk (B), HASTE sequence.

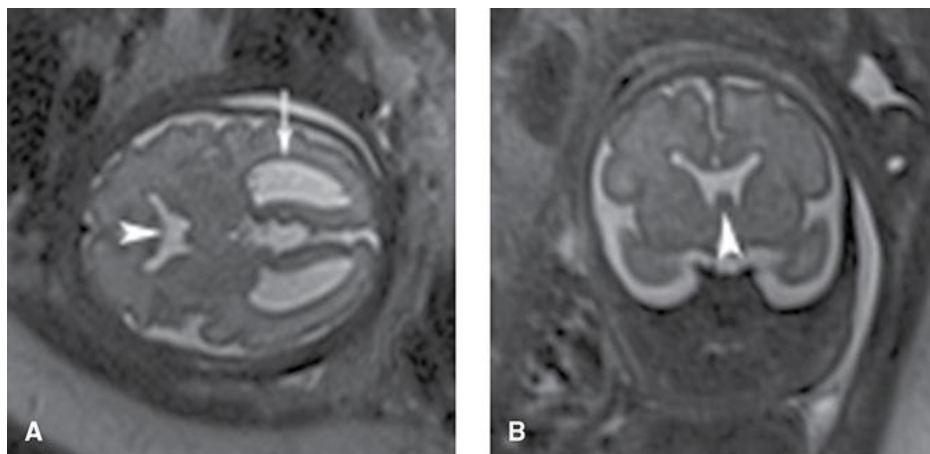


Figure 14-14. Septo-optic dysplasia at 30 wk: absent cavum septum pellucidum with mild ventriculomegaly.

Thorax

- Size and location of masses, **lung volume**; feeding vessel distinguishes **CPAM vs pulmonary sequestration**.
- **CDH:** quantifies **herniated liver** and compressed lung.
- Lung volumes in prolonged oligohydramnios (renal anomalies, ruptured membranes).

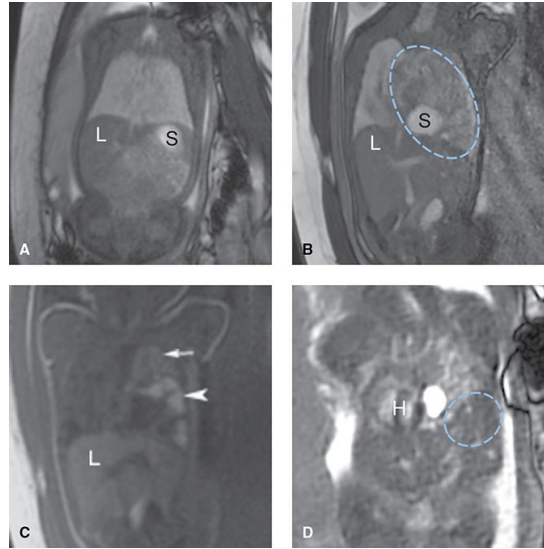


Figure 14-15. Normal lungs (A) vs left CDH with stomach, bowel and meconium-filled colon in the chest and heart shifted right (B–D).

Abdomen

- **Meconium is T1-bright** → GI anomalies and **cloacal malformations**; distinguishes colon (meconium) from bladder (urine).
- **Meconium peritonitis calcifications: ultrasound is better.** Pseudocysts and abnormal meconium migration: MR is better.

Adjunct to fetal therapy

- **Myelomeningocele** (brain/spine before repair); **sacroccoccygeal teratoma** (pelvic extension).
- **Neck masses** → extent and airway compression → identifies candidates for **EXIT**. **Jaw index** for severe micrognathia.
- Before **laser for TTTS**: detect ICH or periventricular leukomalacia.

Placenta

- Complements ultrasound in **PAS**, especially to define the **extent** of invasion. High sensitivity when combined with ultrasound; findings correlate with the need for **cesarean hysterectomy**.
- **MR signs of PAS: dark intraplacental T2 bands, focal bulge**, placental **heterogeneity**, involvement of the bladder–serosal interface, fibrin deposition.
- Emerging: **arterial spin labelling** (maternal RBCs as endogenous contrast, perfusion) and **radiomics texture analysis**.

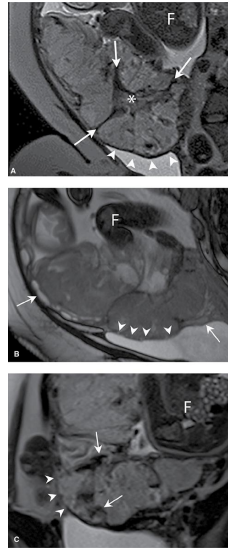


Figure 14-16. Placenta percreta on MR: dark T2 bands, a bulge at the bladder–serosal interface and marked heterogeneity.

Quick-Recall: Which Test Answers Which Question

Clinical question	Best test / sign	Key threshold
Dating	CRL (<14 wk)	Redate if >5 d (<9 wk) / >7 d (9–<14 wk)
Anembryonic gestation	Transvaginal MSD	≥25 mm with no embryo
Embryonic demise	Transvaginal CRL	No heartbeat at CRL ≥7 mm (Parkland ≥10 mm)
Aneuploidy/cardiac risk	Nuchal translucency (11–14 wk)	≥3 mm → detailed scan; ≥3.5 mm → fetal echo
Previa / low-lying	Transvaginal scan (≥16 wk)	Covers/reaches os / within 2 cm ; rescan 32 wk
Vasa previa	TV color + pulsed Doppler	Fetal vessels within 2 cm of os; fetal HR on Doppler
PAS	US (5 signs) ± MR	Myometrium <1 mm , lacunae, bridging vessels, bulge
Preterm birth risk	Transvaginal cervical length (≥16 wk)	Shortest of 3; measure distal to funnel
Oligohydramnios	SDP / AFI	SDP <2 cm / AFI <5 cm
Hydramnios	SDP / AFI	SDP ≥8 cm / AFI ≥24–25 cm ; severe AFI ≥35 / SDP ≥12
FGR management	Umbilical artery Doppler	S/D >95th percentile; AEDF/REDF
Fetal anemia	MCA PSV	≥1.5 MoM
NSAID exposure	Ductus arteriosus Doppler	Use >72 h; 15× constriction odds
Screening for preeclampsia/FGR	Uterine artery Doppler	Not recommended
Complex CNS/thoracic/GI anomaly, PAS extent	MRI (no gadolinium)	Best after 24 wk for CNS

Key Numbers to Memorize

- Transducers: TV **5–12 MHz**, TA **4–6 MHz**, late/obese **2–5 MHz**; TIs **<10 wk**, TIb **≥10 wk**; 1st-trimester Doppler **TI ≤0.7**
- CRL accuracy **±5–7 d**; biometry before 22 wk **±7–10 d**; EFW **±15%**; IVF EDD **+266 d** (day-3 frozen **+263 d**)
- Sac **5 wk**, heartbeat **6 wk**; MSD **25 mm** must show embryo; heartbeat by CRL **7 mm**
- Anatomy scan **18–22 wk**; standard detection **~60%**, detailed **>90%**; obesity **↓20%**; anencephaly **98%**

- Previa/low-lying: **2 cm**; follow-up **32 wk**, then **36 wk**; PAS US sensitivity **~90%**, specificity **97–99%**
- Fluid: **30 mL (10 wk) → 200 mL (16 wk) → 800 mL**; urine **1000 mL/d**, swallowing **750 mL/d**, lung fluid **350 mL/d**
- SDP normal **2–8 cm**; AFI normal **5–24/25 cm**; $AFI \approx 3 \times SDP$
- Hydramnios **1–2%**; anomaly **~15%**, diabetes **15–20%**, idiopathic **up to 70%**; anomalous neonate **8 / 12 / >30%** (mild/moderate/severe)
- Amnioreduction **1000–2000 mL over 20–30 min**, 18–20 G needle
- Oligohydramnios: deliver **36⁰–37⁶ wk**; major malformation **25%** with $AFI \leq 5$ at 24–34 wk
- UA S/D: **≤ 4 (>20 wk), <3 (>30 wk), ~2 (term)**; REDF = **>70%** villous artery obliteration
- MCA PSV **1.5 MoM**; angle correction **$\leq 30^\circ$** ; CPR **<1** → **11×** adverse outcome
- MRI: **1.5–3 T**; CNS diagnosis changed in **~50%**, management in **20%**; no gadolinium unless life-threatening